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Course Name: Pattern Recognition1

Assignment1

2. Dataset Description

The dataset includes observations of three penguin species:

Adelie

Chinstrap

Gentoo

Each observation contains several biological measurements:

Bill length (mm)

Bill depth (mm)

Flipper length (mm)

Body mass (g)

Sex

Island

Species (target variable)

The dataset initially contained missing values, which were removed to ensure reliability of the analysis. After preprocessing, the dataset was clean and suitable for statistical evaluation.

3. Data Preparation

The preprocessing phase included several essential steps:

Loading the dataset using Pandas

Inspecting structure using shape, info, and describe

Identifying and removing missing values using dropna()

Converting categorical variables (species, sex, island) into categorical data types

These steps ensured data consistency and prepared the dataset for visualization and statistical testing. Removing missing values was necessary to avoid bias in the results.

4. Exploratory Data Analysis (EDA)

EDA was conducted using multiple visualization techniques to understand the structure and relationships within the dataset.

Key visual methods:

Histograms to analyze feature distributions

Pairplots to examine relationships between variables

Boxplots to compare species across key features

Observations:

The Gentoo species is clearly distinguishable from the other species, particularly in flipper length and body mass.

Adelie and Chinstrap show partial overlap, indicating similarity in some biological traits.

Among all features, flipper length and body mass appear to provide the strongest separation between species.

These observations suggest that certain morphological features are more informative for classification than others.

5. Hypothesis Testing

5.1 One-Way ANOVA Test

ANOVA was used to determine whether there are significant differences in flipper length among the three penguin species.

Hypotheses:

H0: The mean flipper length is equal across all species

H1: At least one species has a different mean flipper length

Result:

The test produced a very small p-value ($p < 0.05$), leading to rejection of the null hypothesis.

Interpretation:

This indicates that flipper length differs significantly between at least one pair of species. This confirms what was visually observed during EDA.

5.2 Chi-Square Test of Independence

Chi-Square Test was applied to examine the relationship between species and body mass categories.

Procedure:

Body mass was categorized into Light, Medium, and Heavy

A contingency table was created

The Chi-square test was applied

Hypotheses:

H0: Species and body mass category are independent

H1: Species and body mass category are dependent

Result:

The p-value was significantly lower than 0.05, leading to rejection of the null hypothesis.

Interpretation:

This confirms a strong statistical association between penguin species and body mass distribution.

6. Results & Discussion

The analysis revealed clear biological and statistical patterns across penguin species. The visualization results showed that Gentoo species is distinctly separated from the other species, while Adelie and Chinstrap show partial overlap.

The statistical tests strongly support these findings:

ANOVA confirmed significant differences in flipper length across species.

Chi-Square test confirmed that body mass category depends on species.

Overall, flipper length and body mass are the most informative features for distinguishing between penguin species, making them highly relevant for classification tasks.

7. Conclusion

This study demonstrates the effectiveness of combining Exploratory Data Analysis with statistical hypothesis testing in understanding real-world datasets.

The main conclusions are:

Significant differences exist between penguin species.

Flipper length and body mass are key discriminative features.

Species classification is strongly supported by statistical evidence.

The results highlight the importance of data-driven analysis in pattern recognition tasks and confirm that meaningful biological differences can be extracted using statistical methods.